

Photosynthesis Bench

Finding out what actually sets the rate of photosynthesis — and what the Calvin cycle is really waiting for when the light goes out.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/photosynthesis-bench/

1 Before you open anything

Answer from what you already know. Getting it right is not the point; having something on paper to argue with later is.

PREDICTION

A C₃ leaf sits at 25 °C in air at 415 ppm CO₂. You raise the light intensity steadily from deep shade (about 150 μmol photons m⁻² s⁻¹) all the way to full midday sun (2000). Describe what the rate of carbon fixation does across that range — does it climb in proportion the whole way, level off, or something else?

If you said it levels off, name the thing you think decides *how high* the level is. If you said it does not, say why not.

2 Set the bench up and read it

Open the demonstration. Press **Reset** first — that returns every control to its starting condition and settles the chloroplast at steady state immediately.

1. Confirm the three controls read **Light intensity 1000 μmol m⁻² s⁻¹**, **CO₂ around the leaf 415 ppm** and **Temperature 25.0 °C**.
2. Fill in the *Lights on* column of the table from the two ceiling bars and the *Live readouts* panel.

3. Now press **Lights out** and keep your eyes on the readouts. Fill in the *Lights out* column from what happens in the first two or three seconds of darkness. The tile labelled *Carbon fixation after the light went out* is counting for you and tells you what percentage of the lit rate is still running.
4. Wait until that tile stops counting and records a final time. Write it in the last row.

Readout, using its on-screen label	Lights on	First 2–3 s after <i>Lights out</i>
Light reactions can supply		
CO ₂ + Rubisco can support		
What is limiting right now		
Carbon fixed (gross)		
O ₂ evolved		
ATP pool (%)		
NADPH pool (%)		
Calvin enzymes active (%)		
Net CO ₂ exchange		
Carbon fixation after the light went out – final time		

a. One of those two rates went to zero the instant the light died and the other did not. Which was which, and what does that difference tell you about where photons are actually consumed?

b. While the light was off, what was paying for the carbon that was still being fixed? Answer in terms of the two pool readouts, not in general terms.

c. *Carbon fixed (gross)* and *Net CO₂ exchange* differed by a fixed amount in the lit column. What is that amount, and what process does it represent?

The pools on this bench hold 450 $\mu\text{mol ATP}$ and 300 $\mu\text{mol NADPH}$ per m^2 , roughly a hundred times what a real chloroplast carries. That is deliberate, and the demonstration says so: it slows the drain down to something you can watch. A real leaf stops fixing carbon within a second or two of the light going out. Do not quote the time you measured as a fact about a real plant — quote the shape of what happened.

3 The light response curve, one point at a time

Press **Reset**, then hold CO_2 at 415 ppm and the temperature at 25.0 °C for the whole of this part. Only the light moves.

1. Set **Light intensity** to the first value in the table. The **Shade**, **Overcast**, **Bright** and **Full sun** buttons give you 150, 600, 1000 and 2000 directly; use the slider for 300 and 1500.
2. Wait about 30 seconds. Watch *Calvin enzymes active* — the reading you want is the one you get after that number has more or less stopped moving.
3. Record the row, then move to the next light and repeat. Work upwards through the table.

Light ($\mu\text{mol photons m}^{-2} \text{ s}^{-1}$)	Light reactions can supply	CO_2 + Rubisco can support	Carbon fixed (gross)	ATP pool (%)	What is limiting right now
150					
300					
600					
1000					
1500					
2000					

a. Compare your *Carbon fixed (gross)* column against the two ceiling columns row by row. State the arithmetic rule that turns the two ceilings into the rate. It is one short sentence, and it is not an average and not a product.

b. Between 150 and 2000 the light ceiling roughly triples but the fixation rate does not. Say in your own words what the light response curve does across this range and what is holding the top of it down.

c. The *ATP pool* column changes character partway down the table. Describe what a nearly full pool tells you about which ceiling is binding, and what a near-empty pool tells you. Then explain the mechanism: why does a pool that is nearly full force the light reactions to slow down?

d. Go back to your prediction in Part 1. Which parts of it held up, and which specific assumption turned out to be incomplete?

The 30-second wait is not the page being slow. Four Calvin enzymes — GAPDH, FBPase, SBPase and PRK — are switched on by light through the ferredoxin–thioredoxin system, and that switching takes tens of seconds in either direction. It is the reason the CO₂ ceiling drifts for a while after you move the light slider even though you never touched the CO₂.

4 Working backwards, and a shortcut that fails

A limiting-factor question rarely hands you the answer. It hands you a change and asks what it will do.

Working backwards. The bench models rubisco's carboxylation capacity as

$ceiling = V_c \times C_i / (C_i + K_m) \times activation$, with $V_c = 40 \mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$ at 25 °C, $K_m = 300 \text{ ppm}$, and $C_i = 0.70 \times C_a$.

1. Press **Reset**. Read *Light reactions can supply* and *Calvin enzymes active*, and write both here: _____ and _____.
2. On paper, work out the C_a (the *CO₂ around the leaf* setting) that would make the CO_2 ceiling equal that light ceiling, leaving light and temperature where they are. Show the algebra.
3. Set the **CO₂ around the leaf** slider to your answer, to the nearest 5 ppm, and read *What is limiting right now*.

a. What did *What is limiting right now* say, and how far was your calculated C_a from making the two ceiling numbers match on screen?

The shortcut. "More CO_2 gives more photosynthesis" is a reasonable-sounding claim. Test it twice, at two different light levels, and record what the same CO_2 move buys you each time.

1. Press **Reset**, then press **Shade** (light 150). Wait for the readings to settle and record *Carbon fixed (gross)*.
2. Drag **CO₂ around the leaf** from 415 to **1200 ppm**. Record *Carbon fixed (gross)* again.
3. Press **Reset**, then **Full sun** (light 2000), wait, and repeat the identical CO_2 move.

Light	Carbon fixed at 415 ppm	Carbon fixed at 1200 ppm	Change	What is limiting right now, at 415 ppm
150 (Shade)				
2000 (Full sun)				

b. The same slider, moved the same distance, in the same leaf. Explain why one row shows a large change and the other shows none, using the word *ceiling*.

c. Rewrite “more CO₂ gives more photosynthesis” as a claim that is actually true, by attaching the condition it needs.

d. Switch the plot to **Rate vs light**. Below a certain light intensity the 400 ppm and 1200 ppm curves lie on top of each other. Read that intensity off the caption and record it: _____ $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$. Which of your two table rows sits below it, and which sits above?

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it, set the stated conditions, and check yourself — and where you were off, write down which step went wrong.

1. A leaf is at $1000 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$, 25°C , and $C_a = 500 \text{ ppm}$. Use $V_c = 40$, $K_m = 300 \text{ ppm}$, $C_i = 0.70 C_a$, and an enzyme activation of $I/(I + 50)$. The light reactions can supply $25.9 \mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$ under these conditions. Find C_i , the CO₂ ceiling, the gross fixation rate, and which input is limiting.

2. A leaf is fixing carbon at a steady $22.0 \mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$. Using $4 \text{ e}^- \rightarrow 1 \text{ O}_2 + 2 \text{ NADPH} + 3 \text{ ATP}$, and $3 \text{ ATP} + 2 \text{ NADPH}$ spent per CO_2 fixed, calculate (i) the rate of O_2 evolution, (ii) the electron flow through the thylakoid, and (iii) the rate at which the Calvin cycle is consuming ATP. Then explain why this stoichiometry forces O_2 out to equal CO_2 in at steady state, and name one real process the bench leaves out that would break that 1:1 ratio in an actual C_3 leaf.

3. The ATP pool on the bench holds $450 \mu\text{mol m}^{-2}$ and sits 93.8% full while the leaf fixes carbon at $18.7 \mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$. The light is switched off, so nothing refills it. (i) Calculate the rate of ATP consumption. (ii) Calculate how long the pool would last if the Calvin cycle kept turning at that same rate until the pool was empty. (iii) The measured time in Part 2 was longer than your answer to (ii). Explain why, referring to what happens to the rate of fixation as the pool falls.

4. Mitochondrial respiration on the bench is $1.2 \mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$ at 25°C with a Q_{10} of 2.0. At 45°C , $1000 \mu\text{mol photons}$ and 415 ppm , gross fixation is $4.64 \mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$. (i) Calculate respiration at 45°C and the net CO_2 exchange, and say what the sign of your answer means the leaf is doing. (ii) At 45°C the light and CO_2 supplies outside the leaf are unchanged. Explain, in terms of the two ceilings, what has actually gone wrong – and which of the two is the heat-sensitive one.

6 On your own

Nothing here is graded, and the bench does considerably more than this worksheet used.

- Take the temperature from 25 to 35 °C at 1000 μmol photons and 415 ppm and watch *What is limiting right now*. It flips. Work out which ceiling rose and which barely moved, and why warming would do that to one enzyme system and not the other.
- Press **Night**, leave it there for two or three minutes, and watch *Calvin enzymes active* rather than the pools. Then press **Lights back on** and time how long the rate takes to recover. What does that lag say about a leaf under a cloud that passes overhead every few seconds?
- On the **Live trace** plot, the NADPH line is drawn as a wide band with the ATP line on top of it. Set any condition you like and see whether you can ever get the two apart. Then read the stoichiometry on the chloroplast diagram and explain why not.
- Read the panel headed *Read this before you trust the numbers*. Pick one thing it says is not modelled and work out what the *Rate vs light* plot would look like if it were.